

CHE 201 Spring 2020 HW 8

Thanh Nguyen

TOTAL POINTS

95 / 100

QUESTION 1

11 10 / 10

- ✓ - **0 pts** Correct
- **1 pts** Wrong a)

QUESTION 2

22 15 / 15

- ✓ - **0 pts** Correct
- **3.5 pts** Wrong a) i
- **4 pts** Wrong a) ii
- **3.5 pts** Wrong b) i
- **4 pts** Wrong b) ii

QUESTION 3

33 15 / 15

- ✓ - **0 pts** Correct
- **1 pts** simplify the answer
- **5 pts** Wrong C
- **5 pts** Wrong B

QUESTION 4

44 15 / 15

- ✓ - **0 pts** Correct
- **3 pts** Wrong a)
- **3 pts** Wrong b)
- **3 pts** Wrong c)
- **3 pts** Wrong d)
- **3 pts** Wrong max efficiency

QUESTION 5

55 15 / 15

- ✓ - **0 pts** Correct

QUESTION 6

66 15 / 15

- ✓ - **0 pts** Correct

- **5 pts** Wrong a)

QUESTION 7

77 10 / 15

- **0 pts** Correct
- ✓ - **2.5 pts** Wrong a)
- **5 pts** Wrong a)
- ✓ - **2.5 pts** Wrong b)
- **5 pts** Wrong b)
- **2.5 pts** Wrong c)
- **5 pts** Wrong c)

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1.) For a closed system:

$$[\text{entropy change}] = [\text{entropy transfer}] + [\text{entropy production}]$$

- a.) $[\text{entropy production}] = [>0] - [0] \Rightarrow \text{entropy production} > 0 \Rightarrow \text{possible}$
b.) $\text{entropy production} > 0 \Rightarrow \text{possible}$
c.) $[\text{entropy production}] = [0] - [>0] \Rightarrow \text{entropy production} < 0 \Rightarrow \text{impossible}$
d.) $[\text{entropy production}] = [>0] - [>0] \Rightarrow \text{indeterminate}$
e.) $[\text{entropy production}] = [0] - [<0] \Rightarrow \text{entropy production} > 0 \Rightarrow \text{possible}$
f.) $\text{entropy production} < 0 \Rightarrow \text{impossible}$
g.) $[\text{entropy production}] = [<0] - [<0] \Rightarrow \text{indeterminate}$

2.) $\eta_R = 3\eta_I$

a.) $Q_H = Q'_H$
(i) $\eta_R = \frac{W_R}{Q_H}$ and $\eta_I = \frac{W_I}{Q'_H} \Rightarrow \frac{W_R}{W_I} = \frac{\eta_R}{\eta_I} = 3 \Rightarrow W_R = 3W_I \Rightarrow W_R > W_I$

(ii) $Q_C = Q_H - W_R$ and $Q'_C = Q'_H - W_I \Rightarrow Q_C < Q'_C$

b.) $W_R = W_I$

(i) $\eta_R = \frac{W_R}{Q_H}$ and $\eta_I = \frac{W_I}{Q'_H} \Rightarrow \frac{Q_H}{Q'_H} = \frac{\eta_I}{\eta_R} = \frac{1}{3} \Rightarrow Q_H = \frac{1}{3}Q'_H \Rightarrow Q_H < Q'_H$

(ii) $Q_C = Q_H - W_R$ and $Q'_C = Q'_H - W_I \Rightarrow Q_C < Q'_C$

11 10 / 10

✓ - 0 pts Correct

- 1 pts Wrong a)

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1.) For a closed system:

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2 2 15 / 15

✓ - 0 pts Correct

- 3.5 pts Wrong a) i

- 4 pts Wrong a)ii

- 3.5 pts Wrong b)i

- 4 pts Wrong b)ii

3.)

$$a.) \eta_1 = \frac{W_1}{Q_H} = \frac{Q_H - Q}{Q_H} = 1 - \frac{Q}{Q_H}$$

$$\eta_2 = \frac{W_2}{Q} = \frac{Q - Q_c}{Q} = 1 - \frac{Q_c}{Q}$$

$$\eta_{12} = \frac{W_{total}}{Q_H} = \frac{W_1 + W_2}{Q_H} = \frac{W_1}{Q_H} + \frac{W_2}{Q_H} = \eta_1 + \frac{W_2}{Q} \cdot \frac{Q}{Q_H} = \eta_1 + \eta_2 (1 - \eta_1)$$

$$= \eta_1 + \eta_2 - \eta_1 \eta_2$$

$$b.) \eta_1 = 1 - \frac{T}{T_H}; \eta_2 = 1 - \frac{T_c}{T}$$

$$\Rightarrow \eta_{12} = \left(1 - \frac{T}{T_H}\right) + \left(1 - \frac{T_c}{T}\right) - \left(1 - \frac{T}{T_H}\right)\left(1 - \frac{T_c}{T}\right)$$

$$= 2 - \frac{T}{T_H} - \frac{T_c}{T} - \left(1 - \frac{T_c}{T} - \frac{T}{T_H} + \frac{T_c}{T_H}\right) = 1 - \frac{T_c}{T_H}$$

This reversible overall cycle has the same thermal efficiency as the reversible cycle with the same thermal reservoirs T_H and T_c without the intermediate temperature T . It does not depend on the temperature T .

$$c.) \eta_1 = \eta_2 \Rightarrow 1 - \frac{T}{T_H} = 1 - \frac{T_c}{T} \Rightarrow \frac{T}{T_H} = \frac{T_c}{T} \Rightarrow T^2 = T_c T_H \Rightarrow T = \sqrt{T_c T_H}$$

$$4.) \eta_{max} = \eta_{rev} = 1 - \frac{T_c}{T_H} = 1 - \frac{400K}{1200K} = 0.667$$

$$a.) \eta = \frac{W_{cycle}}{Q_H} = \frac{450kJ}{900kJ} = 0.5 < \eta_{max} \Rightarrow \text{irreversibly}$$

$$b.) \eta = \frac{W_{cycle}}{Q_H} = \frac{Q_H - Q_c}{Q_H} = \frac{900kJ - 300kJ}{900kJ} = 0.667 = \eta_{max} \Rightarrow \text{reversibly}$$

$$c.) \eta = \frac{W_{cycle}}{Q_H} = \frac{W_{cycle}}{W_{cycle} + Q_c} = \frac{600kJ}{600kJ + 400kJ} = 0.6 < \eta_{max} \Rightarrow \text{irreversibly}$$

$$d.) \eta = 75\% = 0.75 > \eta_{max} \Rightarrow \text{impossible}$$

33 15 / 15

✓ - 0 pts Correct

- 1 pts simplify the answer

- 5 pts Wrong C

- 5 pts Wrong B

3.)

$$a.) \eta_1 = \frac{W_1}{Q_H} = \frac{Q_H - Q}{Q_H} = 1 - \frac{Q}{Q_H}$$

$$\eta_2 = \frac{W_2}{Q} = \frac{Q - Q_c}{Q} = 1 - \frac{Q_c}{Q}$$

$$\eta_{12} = \frac{W_{total}}{Q_H} = \frac{W_1 + W_2}{Q_H} = \frac{W_1}{Q_H} + \frac{W_2}{Q_H} = \eta_1 + \frac{W_2}{Q} \cdot \frac{Q}{Q_H} = \eta_1 + \eta_2 (1 - \eta_1)$$

$$= \eta_1 + \eta_2 - \eta_1 \eta_2$$

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This reversible overall cycle has the same thermal efficiency as the reversible cycle with the same thermal reservoirs T_H and T_c without the intermediate temperature T . It does not depend on the temperature T .

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$$4.) \eta_{max} = \eta_{rev} = 1 - \frac{T_c}{T_H} = 1 - \frac{400K}{1200K} = 0.667$$

$$a.) \eta = \frac{W_{cycle}}{Q_H} = \frac{450kJ}{900kJ} = 0.5 < \eta_{max} \Rightarrow \text{irreversibly}$$

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$$d.) \eta = 75\% = 0.75 > \eta_{max} \Rightarrow \text{impossible}$$

4 4 15 / 15

✓ - 0 pts Correct

- 3 pts Wrong a)
- 3 pts Wrong b)
- 3 pts Wrong c)
- 3 pts Wrong d)
- 3 pts Wrong max efficiency

5.)

$$a.) W_1 = W_2 \Rightarrow Q_H - Q = Q - Q_c \Rightarrow Q = \frac{Q_H + Q_c}{2}$$

$$\text{For a reversible cycle, } \frac{Q}{Q_H} = \frac{T}{T_H} \Rightarrow Q_H = \frac{T_H}{T} Q ; \frac{Q}{Q_c} = \frac{T}{T_c} \Rightarrow Q_c = \frac{T_c}{T} Q$$

$$\Rightarrow Q = \frac{1}{2} \left(\frac{T_H}{T} Q + \frac{T_c}{T} Q \right) \Rightarrow T = \frac{1}{2} (T_H + T_c) = \frac{1}{2} (1000^\circ R + 400^\circ R) = 700^\circ R$$

$$\eta_1 = 1 - \frac{T}{T_H} = 1 - \frac{700^\circ R}{1000^\circ R} = 0.3 ; \eta_2 = 1 - \frac{T_c}{T} = 1 - \frac{400^\circ R}{700^\circ R} = 0.429$$

$$b.) \eta_s = 1 - \frac{T_c}{T_H} = 1 - \frac{400^\circ R}{1000^\circ R} = 0.6$$

$$W_s = Q_H - Q_c = (Q_H - Q) + (Q - Q_c) = 2W_{\text{cycle}}$$

$$6.) \dot{Q}_H = 120,000 \text{ kJ/h}$$

a.) For a period of 14 days

$$Q_H = 120,000 \text{ kJ/h} \times \frac{24 \text{ h}}{1 \text{ day}} \times 14 \text{ days} = 4.032 \times 10^7 \text{ kJ}$$

$$W_{\text{cycle}} = 1490 \text{ kWh} \times \frac{3600 \text{ s}}{1 \text{ h}} = 5.364 \times 10^6 \text{ kJ}$$

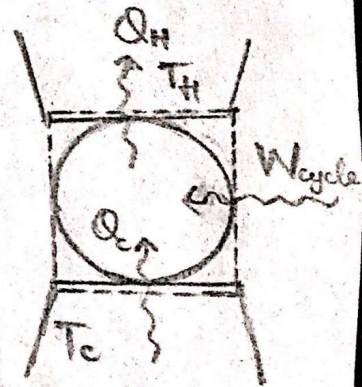
$$\Rightarrow Q_c = Q_H - W_{\text{cycle}} = 4.032 \times 10^7 \text{ kJ} - 5.364 \times 10^6 \text{ kJ} = 3.4956 \times 10^7 \text{ kJ}$$

$$b.) \gamma = \frac{Q_H}{W_{\text{cycle}}} = \frac{4.032 \times 10^7 \text{ kJ}}{5.364 \times 10^6 \text{ kJ}} = 7.517$$

$$c.) T_H = 21^\circ C = 21^\circ C + 273 = 294 \text{ K}$$

$$T_c = 9^\circ C = 9^\circ C + 273 = 282 \text{ K}$$

$$\gamma_{\text{rev}} = \frac{T_H}{T_H - T_c} = \frac{294 \text{ K}}{294 \text{ K} - 282 \text{ K}} = 24.5$$



5 5 15 / 15

✓ - 0 pts Correct

5.)

$$a.) W_1 = W_2 \Rightarrow Q_H - Q = Q - Q_c \Rightarrow Q = \frac{Q_H + Q_c}{2}$$

$$\text{For a reversible cycle, } \frac{Q}{Q_H} = \frac{T}{T_H} \Rightarrow Q_H = \frac{T_H}{T} Q ; \frac{Q}{Q_c} = \frac{T}{T_c} \Rightarrow Q_c = \frac{T_c}{T} Q$$

$$\Rightarrow Q = \frac{1}{2} \left(\frac{T_H}{T} Q + \frac{T_c}{T} Q \right) \Rightarrow T = \frac{1}{2} (T_H + T_c) = \frac{1}{2} (1000^\circ R + 400^\circ R) = 700^\circ R$$

$$\eta_1 = 1 - \frac{T}{T_H} = 1 - \frac{700^\circ R}{1000^\circ R} = 0.3 ; \eta_2 = 1 - \frac{T_c}{T} = 1 - \frac{400^\circ R}{700^\circ R} = 0.429$$

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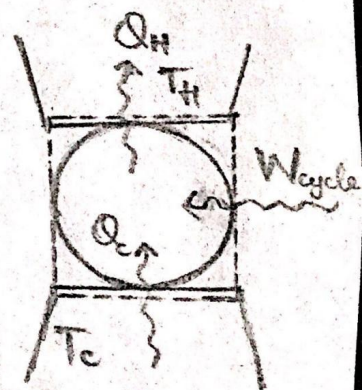
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$$c.) T_H = 21^\circ C = 21^\circ C + 273 = 294 \text{ K}$$

$$T_c = 9^\circ C = 9^\circ C + 273 = 282 \text{ K}$$

$$\gamma_{\text{rev}} = \frac{T_H}{T_H - T_c} = \frac{294 \text{ K}}{294 \text{ K} - 282 \text{ K}} = 24.5$$



6 6 15 / 15

✓ - 0 pts Correct

- 5 pts Wrong a)

7.) Engineering Model

1. Steady state
2. $\Delta KE = \Delta PE = 0$

Using table A-3

•) At $p = 1 \text{ MPa} = 10 \text{ bar}$, $T_H = 179.9^\circ\text{C} = 452.9 \text{ K}$

$$v_4 = 1.1273 \times 10^{-3} \text{ m}^3/\text{kg} ; h_4 = 762.81 \text{ kJ/kg}$$

$$v_1 = 0.1944 \text{ m}^3/\text{kg} ; h_1 = 2778.1 \text{ kJ/kg}$$

•) At $p = 20 \text{ kPa} = 0.2 \text{ bar}$, $T_C = 60.06^\circ\text{C} = 333.06 \text{ K}$

$$v_2 = 1.0172 \times 10^{-3} \text{ m}^3/\text{kg} ; h_2 = 251.40 \text{ kJ/kg}$$

$$v_3 = 7.649 \text{ m}^3/\text{kg} ; h_3 = 2609.7 \text{ kJ/kg}$$

Using the equation $v = (1-x)v_f + xv_g$ and $h = (1-x)h_f + xh_g$

•) With $x = 0.88 \rightarrow v_2 = 6.731 \text{ m}^3/\text{kg} ; h_2 = 2326.704 \text{ kJ/kg}$

•) With $x = 0.18 \rightarrow v_3 = 1.378 \text{ m}^3/\text{kg} ; h_3 = 675.894 \text{ kJ/kg}$

Energy rate balance

$$\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2) = 0 \rightarrow \dot{Q}_{cv} = \dot{W}_{cv} - \dot{m}(h_1 - h_2)$$

a.) Process 2-3:

$$\begin{aligned} \dot{Q}_{23} &= \dot{W}_{23} - \dot{m}(h_2 - h_3) = \dot{m}p(v_3 - v_2) - \dot{m}(h_2 - h_3) = \dot{m}[p(v_3 - v_2) - (h_2 - h_3)] \\ &= \dot{m}[(20 \text{ kPa})(1.378 - 6.731) \text{ m}^3/\text{kg} - (2326.704 - 675.894) \text{ kJ/kg}] = (-1757.9 \text{ kJ/kg}) \dot{m} \end{aligned}$$

Process 4-1:

$$\begin{aligned} \dot{Q}_{41} &= \dot{m}[p(v_1 - v_4) - (h_1 - h_4)] \\ &= \dot{m}[(10^3 \text{ kPa})(0.1944 - 1.1273 \times 10^{-3}) \text{ m}^3/\text{kg} - (762.81 - 2778.1) \text{ kJ/kg}] = (2208.6 \text{ kJ/kg}) \dot{m} \end{aligned}$$

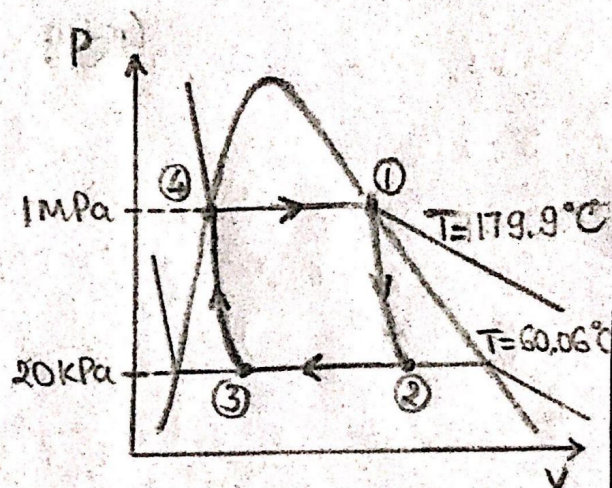
$$-\dot{\sigma}_{\text{cycle}} = \frac{\dot{Q}_{\text{in}}}{T_{\text{in}}} - \frac{\dot{Q}_{\text{out}}}{T_{\text{out}}} = \frac{(2208.6 \text{ kJ/kg}) \dot{m}}{452.9 \text{ K}} - \frac{(1757.9 \text{ kJ/kg}) \dot{m}}{333.06 \text{ K}} = (-0.399 \text{ kJ/kg} \cdot \text{K}) \dot{m}$$

Since $\dot{m} > 0 \rightarrow -\dot{\sigma}_{\text{cycle}} < 0 \rightarrow \dot{\sigma}_{\text{cycle}} > 0 \rightarrow \text{irreversible}$

$$b.) \eta = \frac{\dot{W}_{\text{cycle}}}{\dot{Q}_{\text{in}}} = \frac{\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}}{\dot{Q}_{\text{in}}} = \frac{\dot{Q}_{41} - \dot{Q}_{23}}{\dot{Q}_{41}} = \frac{(2208.6 \text{ kJ/kg}) \dot{m} - (-1757.9 \text{ kJ/kg}) \dot{m}}{(2208.6 \text{ kJ/kg}) \dot{m}} = 0.204$$

$$c.) \eta_{\text{rev}} = 1 - \frac{T_C}{T_H} = 1 - \frac{333.06 \text{ K}}{452.9 \text{ K}} = 0.265$$

$$\eta_{\text{actual}} < \eta_{\text{rev}}$$



77 10 / 15

- 0 pts Correct

✓ - 2.5 pts Wrong a)

- 5 pts Wrong a)

✓ - 2.5 pts Wrong b)

- 5 pts Wrong b)

- 2.5 pts Wrong c)

- 5 pts Wrong c)